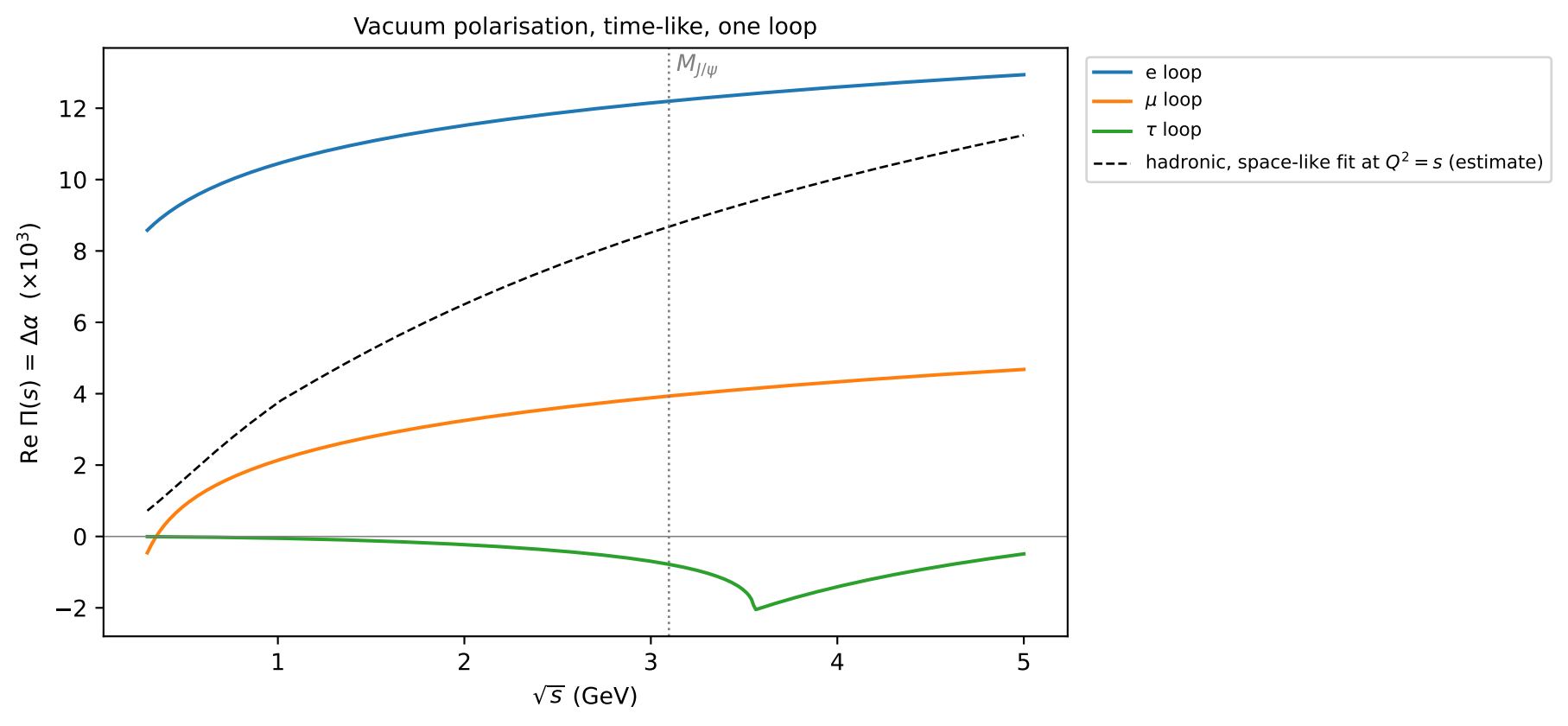
Vacuum polarisation, time-like, one loop
$M_{J/\psi}$
e loop
μ loop
τ loop
hadronic, space-like fit at $Q^2 = s$ (estimate)
Re $\Pi(s) = \Delta\alpha$ ($\times 10^3$)
$\sqrt{s}$ (GeV)
12
10
8
6
4
2
0
−2
1
2
3
4
5